

Top Quark Signal in the Cosmological Higgs Collider

1 Top Quark Signal

We calculate the top-quark contribution to the connected Higgs four-point function in the collapsed, or s -channel squeezed, limit. As in the gauge-boson calculation, the Higgs is treated as massless during inflation. Its Schwinger–Keldysh bulk-to-boundary propagator is:

$$G_a(k; \tau) = \frac{H^2}{2k^3} (1 - iak\tau) e^{iak\tau}, \quad (1)$$

and the scale factor is $a(\tau) = -(H\tau)^{-1}$.

In unitary gauge the top Yukawa interaction is

$$\mathcal{L}_t = it^\dagger \bar{\sigma}^\mu \partial_\mu t + it^{c\dagger} \bar{\sigma}^\mu \partial_\mu t^c - a(\tau) [m_t + g_t \delta h] (tt^c + t^\dagger t^{c\dagger}), \quad (2)$$

$$g_t \equiv \frac{y_t}{\sqrt{2}}, \quad m_t = g_t h_0, \quad \tilde{\nu} \equiv \frac{m_t}{H}. \quad (3)$$

Here t and t^c are two left-handed Weyl fields and have already been conformally rescaled by $a^{3/2}$. Hence every Yukawa vertex carries one power of $a(\tau)$, rather than $a^4(\tau)$.

The fermion formalism of Ref. [2] is written for a two-component Majorana field. It can be applied to the top without changing the physics by the unitary Takagi rotation

$$t = \frac{N_1 + iN_2}{\sqrt{2}}, \quad t^c = \frac{N_1 - iN_2}{\sqrt{2}}. \quad (4)$$

Because the contraction of two Weyl spinors is symmetric, $N_1 N_2 = N_2 N_1$, this gives

$$tt^c = \frac{1}{2} (N_1 N_1 + N_2 N_2). \quad (5)$$

Thus Eq. (2) becomes

$$\mathcal{L}_t = \sum_{A=1}^2 \left[iN_A^\dagger \bar{\sigma}^\mu \partial_\mu N_A - \frac{am_t}{2} (N_A N_A + N_A^\dagger N_A^\dagger) - \frac{ag_t}{2} \delta h (N_A N_A + N_A^\dagger N_A^\dagger) \right]. \quad (6)$$

Both N_A therefore have the same positive Majorana mass and the same Higgs coupling. This is preferable to diagonalizing the mass matrix with eigenvalues $\pm m_t$: the phase in Eq. (4) puts the result directly in the positive-mass convention used in Ref. [2].

There is no additional factor of two in a closed top loop. A single Majorana functional determinant is one half of a Dirac determinant, while the sum over $A = 1, 2$ supplies the opposite

factor two. The net multiplicity is consequently the standard color factor $N_c = 3$, together with the usual minus sign for a closed fermion loop.

For momentum $\mathbf{k} = k(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, we use the helicity eigenspinors

$$h_+(\theta, \phi) = \begin{pmatrix} e^{-i\phi/2} \cos(\theta/2) \\ e^{i\phi/2} \sin(\theta/2) \end{pmatrix}, \quad h_-(\theta, \phi) = \begin{pmatrix} -e^{-i\phi/2} \sin(\theta/2) \\ e^{i\phi/2} \cos(\theta/2) \end{pmatrix}, \quad (7)$$

The mode functions:

$$u_+(k, \tau) = \frac{\tilde{\nu}}{\sqrt{-2k\tau}} W_{-1/2, i\tilde{\nu}}(2ik\tau), \quad u_-(k, \tau) = \frac{1}{\sqrt{-2k\tau}} W_{1/2, i\tilde{\nu}}(2ik\tau), \quad (8)$$

$$v_+(k, \tau) = \frac{1}{\sqrt{-2k\tau}} W_{1/2, i\tilde{\nu}}(2ik\tau), \quad v_-(k, \tau) = \frac{\tilde{\nu}}{\sqrt{-2k\tau}} W_{-1/2, i\tilde{\nu}}(2ik\tau). \quad (9)$$

There are three useful two-component propagators,

$$D_{ab\alpha\dot{\beta}} \equiv \langle N_\alpha N_{\dot{\beta}}^\dagger \rangle_{ab}, \quad \widehat{D}_{ab\alpha\beta} \equiv \langle N_\alpha N_\beta \rangle_{ab}, \quad \check{D}_{ab\dot{\alpha}\dot{\beta}} \equiv \langle N_{\dot{\alpha}}^\dagger N_{\dot{\beta}}^\dagger \rangle_{ab}. \quad (10)$$

Following Ref. [2], we call D , $\widehat{D}$, and $\check{D}$ the type-1, type-2, and type-3 propagators, respectively. The nonlocal late-time parts of the symmetric type-2 and type-3 propagators are especially simple:

$$\left[\widehat{D}^{(S)\alpha}_{\beta}(k; \tau_1, \tau_2) \right]_{NL\text{oc}} = \sum_{f=\pm} \kappa_f(\tilde{\nu}) (4k^2 \tau_1 \tau_2)^{if\tilde{\nu}} \delta^\alpha_\beta, \quad (11)$$

$$\left[\check{D}^{(S)\dot{\alpha}}_{\dot{\beta}}(k; \tau_1, \tau_2) \right]_{NL\text{oc}} = \sum_{f=\pm} \kappa_f(\tilde{\nu}) (4k^2 \tau_1 \tau_2)^{if\tilde{\nu}} \delta^{\dot{\alpha}}_{\dot{\beta}}, \quad (12)$$

where

$$\kappa_f(\tilde{\nu}) = \frac{\tilde{\nu} \Gamma^2(-2if\tilde{\nu})}{\Gamma(1 - if\tilde{\nu}) \Gamma(-if\tilde{\nu})}, \quad \kappa_- = \kappa_+^*. \quad (13)$$

At zero chemical potential the helicity tensor in the unraised propagator reduces to the invariant antisymmetric tensor,

$$h_{+\alpha} h_{-\beta} - h_{-\alpha} h_{+\beta} = \epsilon_{\alpha\beta}. \quad (14)$$

It follows that the two Wightman orderings have the same type-2 nonlocal part.

The type-1 propagator behaves differently. At zero chemical potential its symmetric nonlocal part vanishes, while its antisymmetric part is proportional to $\sigma \cdot \hat{\mathbf{k}}$. Consequently a pair of type-1 soft lines is not covered by the factorized symmetric prescription of Ref. [2]. We return to this point in Sec. 3.

For one cyclic ordering of the external Higgs legs, the box has the schematic SK form

$$\langle \delta h^4 \rangle'_{\text{Box}} = -N_c (ig_t)^4 \sum_{a,b,c,d=\pm} (abcd) \int \prod_{j=1}^4 d\tau_j a(\tau_j) G_{a_j}(k_j; \tau_j) \int \frac{d^3 \mathbf{q}}{(2\pi)^3} \mathcal{S}_{abcd}, \quad (15)$$

where $(a_1, a_2, a_3, a_4) = (a, b, c, d)$. We now spell out the loop integrand $\mathcal{S}_{abcd}$. Introduce a composite two-component index A , which is undotted for $r = N$ and dotted for $r = \bar{N}$, and write

$$(E_N)^{AB} = \epsilon^{\alpha\beta}, \quad (E_{\bar{N}})^{AB} = \epsilon^{\dot{\alpha}\dot{\beta}}. \quad (16)$$

The propagator joining two vertices is

$$\begin{aligned}\mathcal{D}_{ab,\alpha\beta}^{NN} &= \widehat{D}_{ab,\alpha\beta}, & \mathcal{D}_{ab,\alpha\dot{\beta}}^{N\bar{N}} &= D_{ab,\alpha\dot{\beta}}, \\ \mathcal{D}_{ab,\dot{\alpha}\beta}^{\bar{N}N} &= \bar{D}_{ab,\dot{\alpha}\beta}, & \mathcal{D}_{ab,\dot{\alpha}\dot{\beta}}^{\bar{N}\bar{N}} &= \check{D}_{ab,\dot{\alpha}\dot{\beta}}.\end{aligned}\tag{17}$$

The reversed type-1 propagator is fixed by

$$\bar{D}_{ab,\dot{\alpha}\beta}(\mathbf{p}; \tau, \tau') \equiv -D_{ba,\beta\dot{\alpha}}(-\mathbf{p}; \tau', \tau).\tag{18}$$

We orient the loop as $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$ and choose

$$\mathbf{p}_{12} = \mathbf{q} + \mathbf{k}_1, \quad \mathbf{p}_{23} = \mathbf{K} = \mathbf{q} + \mathbf{k}_s, \quad \mathbf{p}_{34} = \mathbf{q} + \mathbf{k}_3, \quad \mathbf{p}_{41} = \mathbf{q}.\tag{19}$$

The complete integrand in Eq. (15) is then

$$\begin{aligned}\mathcal{S}_{abcd} &= \sum_{r_1, r_2, r_3, r_4=N, \bar{N}} (E_{r_1})^{B_1 A_1} (E_{r_2})^{B_2 A_2} (E_{r_3})^{B_3 A_3} (E_{r_4})^{B_4 A_4} \\ &\quad \times \mathcal{D}_{ab, A_1 B_2}^{r_1 r_2}(\mathbf{p}_{12}; \tau_1, \tau_2) \mathcal{D}_{bc, A_2 B_3}^{r_2 r_3}(\mathbf{p}_{23}; \tau_2, \tau_3) \\ &\quad \times \mathcal{D}_{cd, A_3 B_4}^{r_3 r_4}(\mathbf{p}_{34}; \tau_3, \tau_4) \mathcal{D}_{da, A_4 B_1}^{r_4 r_1}(\mathbf{p}_{41}; \tau_4, \tau_1).\end{aligned}\tag{20}$$

The tensors E_{r_i} are precisely the contractions from the four scalar Yukawa vertices. The factor $1/2$ in the Majorana interaction in Eq. (6) is canceled at each vertex by the two ways of differentiating $N_A N_A$ or $N_A^\dagger N_A^\dagger$; hence no extra numerical factor is missing from Eq. (15).

For example, the first word gives

$$\begin{aligned}\mathcal{S}_{abcd}^{NNNN} &= \epsilon^{\beta_1 \alpha_1} \epsilon^{\beta_2 \alpha_2} \epsilon^{\beta_3 \alpha_3} \epsilon^{\beta_4 \alpha_4} \widehat{D}_{ab, \alpha_1 \beta_2}(\mathbf{p}_{12}) \widehat{D}_{bc, \alpha_2 \beta_3}(\mathbf{p}_{23}) \\ &\quad \times \widehat{D}_{cd, \alpha_3 \beta_4}(\mathbf{p}_{34}) \widehat{D}_{da, \alpha_4 \beta_1}(\mathbf{p}_{41}),\end{aligned}\tag{21}$$

A word containing all four propagator types, row 7, is

$$\begin{aligned}\mathcal{S}_{abcd}^{N\bar{N}\bar{N}N} &= \epsilon^{\beta_1 \alpha_1} \epsilon^{\dot{\beta}_2 \dot{\alpha}_2} \epsilon^{\dot{\beta}_3 \dot{\alpha}_3} \epsilon^{\beta_4 \alpha_4} D_{ab, \alpha_1 \dot{\beta}_2}(\mathbf{p}_{12}) \check{D}_{bc, \dot{\alpha}_2 \dot{\beta}_3}(\mathbf{p}_{23}) \\ &\quad \times \bar{D}_{cd, \dot{\alpha}_3 \beta_4}(\mathbf{p}_{34}) \widehat{D}_{da, \alpha_4 \beta_1}(\mathbf{p}_{41}).\end{aligned}\tag{22}$$

The nonlocal s -channel signal arises when the lines with momenta $\mathbf{q}$ and $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$ are soft. At every Yukawa vertex one may choose either NN or $N^\dagger N^\dagger$. We denote these choices by N and $\bar{N}$. They give $2^4 = 16$ Weyl contractions for a fixed cyclic ordering of the four external Higgs legs. Equation (18) fixes the Fermi sign and the SK/time reversal for every $\bar{D}$ entry. The hard edges are (12) and (34), and the soft edges are (23) and (41). The complete list is

No.	$(r_1 r_2 r_3 r_4)$	$(12, 23, 34, 41)$	soft class
1	$NNNN$	$(\widehat{D}, \widehat{D}, \widehat{D}, \widehat{D})$	AA
2	$NNN\bar{N}$	$(\widehat{D}, \widehat{D}, D, \bar{D})$	AN
3	$NN\bar{N}N$	$(\widehat{D}, D, \bar{D}, \widehat{D})$	NA
4	$NN\bar{N}\bar{N}$	$(\widehat{D}, D, \check{D}, \bar{D})$	NN
5	$N\bar{N}NN$	$(D, \bar{D}, \widehat{D}, \widehat{D})$	NA
6	$N\bar{N}N\bar{N}$	$(D, \bar{D}, D, \bar{D})$	NN
7	$N\bar{N}\bar{N}N$	$(D, \check{D}, \bar{D}, \widehat{D})$	AA
8	$N\bar{N}\bar{N}\bar{N}$	$(D, \check{D}, \check{D}, \bar{D})$	AN
9	$\bar{N}NNN$	$(\bar{D}, \widehat{D}, \widehat{D}, D)$	AN
10	$\bar{N}NN\bar{N}$	$(\bar{D}, \widehat{D}, D, \check{D})$	AA
11	$\bar{N}N\bar{N}N$	$(\bar{D}, D, \bar{D}, D)$	NN
12	$\bar{N}N\bar{N}\bar{N}$	$(\bar{D}, D, \check{D}, \check{D})$	NA
13	$\bar{N}\bar{N}NN$	$(\check{D}, \bar{D}, \widehat{D}, D)$	NN
14	$\bar{N}\bar{N}N\bar{N}$	$(\check{D}, \bar{D}, D, \check{D})$	NA
15	$\bar{N}\bar{N}\bar{N}N$	$(\check{D}, \check{D}, \bar{D}, D)$	AN
16	$\bar{N}\bar{N}\bar{N}\bar{N}$	$(\check{D}, \check{D}, \check{D}, \check{D})$	AA

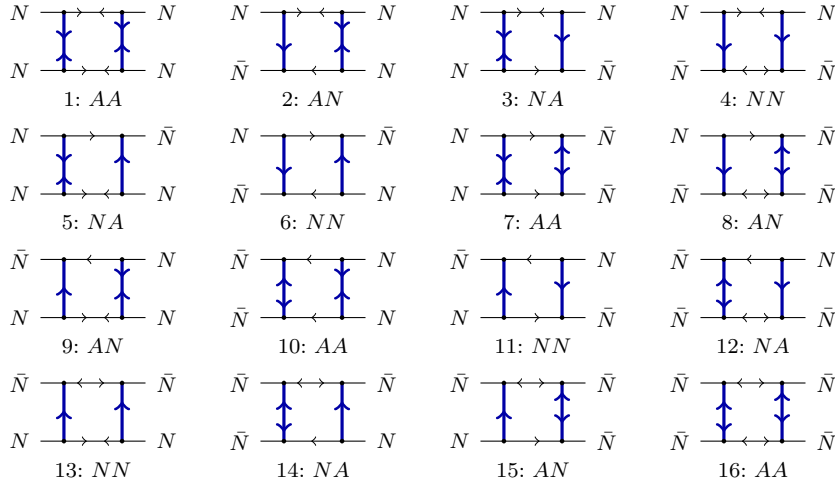


Figure 1: The sixteen Weyl contractions for one cyclic box, using the fermion-arrow convention of Ref. [2]. A type-1 line has one arrow directed from N to $N^\dagger$; the two arrows on a type-2 line point inward, while those on a type-3 line point outward. The horizontal edges (12) and (34) are hard; the blue vertical edges (23) and (41) are soft. The label below each diagram is its soft class.

Here A is only a compact class label for a type-2 $\widehat{D}$ or type-3 $\check{D}$ soft line, while N denotes a type-1 D or $\bar{D}$ soft line.

2 Symmetric soft-line contribution

We first replace both soft lines by their symmetric nonlocal parts. At zero chemical potential,

$$[D^{(S)}]_{NLoc} = 0, \quad [\widehat{D}^{(S)}]_{NLoc} \neq 0, \quad [\check{D}^{(S)}]_{NLoc} \neq 0. \quad (23)$$

Consequently the symmetric nonlocal contributions of rows $\ell \in \{2, 3, 4, 5, 6, 8, 9, 11, 12, 13, 14, 15\}$ vanish, and only

$$\mathcal{G}_{AA} = \{1, 7, 10, 16\} \quad (24)$$

survive. This condition concerns the *soft* edges only: the hard edges of rows 1, 7, 10, and 16 are respectively (2, 2), (1, 1), (1, 1), and (3, 3).

2.1 Loop momentum integral and bubble subgraph

Following the gauge-boson calculation, we keep the two hard propagators unevaluated and perform the soft loop momentum first. For a surviving word $w_\ell = (r_1 r_2 r_3 r_4)$, define

$$\mathcal{H}_{\ell,abcd}(\tau_i) = \text{tr}_2[\mathcal{D}_{ab}^{r_1 r_2}(\mathbf{k}_1; \tau_1, \tau_2) \Delta^{r_2 r_3} \mathcal{D}_{cd}^{r_3 r_4}(\mathbf{k}_3; \tau_3, \tau_4) \Delta^{r_4 r_1}], \quad (25)$$

where the raised-index type-2/type-3 soft matrices are $\Delta^{NN} = \Delta^{\bar{N}\bar{N}} = \sigma^0$. Before doing the $\mathbf{q}$ integral,

$$\mathcal{LOOP}_\ell^{(S)} = \sum_{f=\pm} \kappa_f^2 2^{4if\tilde{\nu}} \left[\prod_{j=1}^4 (-\tau_j)^{if\tilde{\nu}} \right] \mathcal{H}_{\ell,abcd}(\tau_i) \int \frac{d^3 \mathbf{q}}{(2\pi)^3} q^{2if\tilde{\nu}} K^{2if\tilde{\nu}}. \quad (26)$$

The loop integral therefore separates into a bubble subgraph and one extra time factor at each vertex:

$$\mathcal{LOOP}_\ell^{(S)} = \sum_{f=\pm} \left[\prod_{j=1}^4 (-\tau_j)^{if\tilde{\nu}} \right] \mathcal{B}_f^{(S)}(k_s) \mathcal{H}_{\ell,abcd}(\tau_i), \quad (27)$$

where

$$\mathcal{B}_f^{(S)}(k_s) = \kappa_f^2 2^{4if\tilde{\nu}} \int \frac{d^3 \mathbf{q}}{(2\pi)^3} q^{2if\tilde{\nu}} K^{2if\tilde{\nu}} \quad (28)$$

$$= \kappa_f^2 2^{4if\tilde{\nu}} \frac{k_s^{3+4if\tilde{\nu}}}{(4\pi)^{3/2}} f_0(-if\tilde{\nu}, -if\tilde{\nu}). \quad (29)$$

In the notation of the gauge-boson calculation,

$$f_0(x, y) = \Gamma \left[\begin{matrix} x + y - \frac{3}{2}, & \frac{3}{2} - x, & \frac{3}{2} - y \\ 3 - x - y, & x, & y \end{matrix} \right]. \quad (30)$$

The factor $2^{4if\tilde{\nu}}$ follows directly from the normalization in Eq. (12).

2.2 Left and right subgraphs

The left and right subgraph contributions:

$$\begin{aligned} \mathcal{T}_{f,\ell}^{(L)} &= (ig_t)^2 \sum_{a,b=\pm} ab \int_{-\infty}^0 d\tau_1 d\tau_2 a(\tau_1) a(\tau_2) G_a(k_1; \tau_1) G_b(k_2; \tau_2) \\ &\quad \times (-\tau_1)^{if\tilde{\nu}} (-\tau_2)^{if\tilde{\nu}} \mathcal{D}_{ab}^{r_1 r_2}(\mathbf{k}_1; \tau_1, \tau_2), \end{aligned} \quad (31a)$$

$$\begin{aligned} \mathcal{T}_{f,\ell}^{(R)} &= (ig_t)^2 \sum_{c,d=\pm} cd \int_{-\infty}^0 d\tau_3 d\tau_4 a(\tau_3) a(\tau_4) G_c(k_3; \tau_3) G_d(k_4; \tau_4) \\ &\quad \times (-\tau_3)^{if\tilde{\nu}} (-\tau_4)^{if\tilde{\nu}} \mathcal{D}_{cd}^{r_3 r_4}(\mathbf{k}_3; \tau_3, \tau_4). \end{aligned} \quad (31b)$$

Inserting the scale factor and the two massless bulk-boundary propagator into the first line gives:

$$\begin{aligned} \mathcal{T}_{f,\ell}^{(L)} &= (ig_t)^2 \frac{H^2}{4k_1^3 k_2^3} \sum_{a,b=\pm} ab \int_{-\infty}^0 d\tau_1 d\tau_2 (-\tau_1)^{p_f} (-\tau_2)^{p_f} \\ &\quad \times (1 - iak_1\tau_1)(1 - ibk_2\tau_2) e^{iak_1\tau_1 + ibk_2\tau_2} \mathcal{D}_{ab}^{r_1 r_2}(\mathbf{k}_1; \tau_1, \tau_2), \quad p_f = -1 + if\tilde{\nu}. \end{aligned} \quad (32)$$

The right subgraph is obtained by $(1, 2, a, b, r_1, r_2) \rightarrow (3, 4, c, d, r_3, r_4)$.

With $z_1 = -k_1\tau_1$ and $z_2 = -k_2\tau_2$, Eq. (32) becomes

$$\begin{aligned} \mathcal{T}_{f,\ell}^{(L)} &= (ig_t)^2 \frac{H^2}{4k_1^3 k_2^3} \left(\frac{k_{12}}{2} \right)^{-2if\tilde{\nu}} \sum_{a,b=\pm} ab \int_0^\infty dz_1 dz_2 z_1^{p_f} z_2^{p_f} (1 + iaz_1)(1 + ibz_2) \\ &\quad \times e^{-iaz_1 - ibz_2} \mathcal{D}_{ab}^{r_1 r_2}(\mathbf{k}_1; z_1, z_2) \end{aligned} \quad (33)$$

The dimensionless propagator on the last line is the one used in Ref. [2]. For endpoint chiralities $r, s \in \{N, \bar{N}\}$ define

$$\mathcal{I}_{ab}^{rs;p_1 p_2}(\hat{\mathbf{k}}) = \int_0^\infty dz_1 dz_2 e^{-iaz_1 - ibz_2} z_1^{p_1} z_2^{p_2} \left[\mathcal{D}_{ab}^{rs}(\hat{\mathbf{k}}; z_1, z_2) \right]_{\text{mixed}}. \quad (34)$$

Here $\mathcal{D}^{NN} = \widehat{D}$, $\mathcal{D}^{N\bar{N}} = D$, $\mathcal{D}^{\bar{N}N} = \bar{D}$, and $\mathcal{D}^{\bar{N}\bar{N}} = \check{D}$. The subscript mixed means that the anomalous propagators are converted from the two-lower-index matrices of Ref. [2] using the Yukawa ϵ tensor. Following Eqs. (2.35) and (2.37) of that reference, the second index of $\widehat{D}$ and the first index of $\check{D}$ are raised:

$$\widehat{D}_\alpha{}^\beta = \epsilon^{\beta\gamma} \widehat{D}_{\alpha\gamma}, \quad \check{D}^{\dot{\alpha}}{}_{\dot{\beta}} = \epsilon^{\dot{\alpha}\dot{\gamma}} \check{D}_{\dot{\gamma}\dot{\beta}}. \quad (35)$$

The type-1 seed matrices already have the required mixed-index form. This is an index conversion, not an additional propagator. The opposite placement of ϵ for the type-2 and type-3 cases is also what makes both symmetric soft propagators produce the same positive Kronecker delta in Eq. (12). Expanding the two massless polynomials in Eq. (33) gives exactly

$$\mathcal{H}_{f;ab}^{rs} = \mathcal{I}_{ab}^{rs;p_f p_f} + ia\mathcal{I}_{ab}^{rs;p_f+1,p_f} + ib\mathcal{I}_{ab}^{rs;p_f,p_f+1} - ab\mathcal{I}_{ab}^{rs;p_f+1,p_f+1}, \quad (36)$$

$$\mathcal{Q}_f^{rs}(\hat{\mathbf{k}}) = \sum_{a,b=\pm} ab \mathcal{H}_{f;ab}^{rs}(\hat{\mathbf{k}}). \quad (37)$$

Therefore the two original time integrals reduce to

$$\mathcal{T}_{f,\ell}^{(L)} = (ig_t)^2 \frac{H^2}{4k_1^3 k_2^3} \left(\frac{k_{12}}{2} \right)^{-2if\tilde{\nu}} \mathcal{Q}_f^{r_1 r_2}(\hat{\mathbf{k}}_1), \quad (38)$$

$$\mathcal{T}_{f,\ell}^{(R)} = (ig_t)^2 \frac{H^2}{4k_3^3 k_4^3} \left(\frac{k_{34}}{2} \right)^{-2if\tilde{\nu}} \mathcal{Q}_f^{r_3 r_4}(\hat{\mathbf{k}}_3). \quad (39)$$

Appendix A gives the type-1, type-2, and type-3 seed matrices required by Eq. (37). Thus no hard time integral remains.

2.3 Rotation and spinor contraction diagram by diagram

For each of the four surviving diagrams we use the same three steps. We first write its left and right hard subgraphs as finite sums of the integral seeds of Ref. [2] with their *original* spinor indices and hard momentum along $\hat{\mathbf{z}}$. We then rotate those tensors to $\hat{\mathbf{k}}_1$ and $\hat{\mathbf{k}}_3$ without raising or lowering an index. Only in the third step do we use the ϵ tensors at the vertices to join them to the two spinor deltas from the symmetric soft bubble.

We display the $f = +$ oscillatory branch; its complex conjugate gives the $f = -$ branch. Throughout this subsection $p = -1 + i\tilde{\nu}$. The axis-frame seed sums below contain the four terms generated by the two massless Higgs polynomials in Eq. (37). The spinor indices are kept in the form in which the corresponding seed is tabulated in Appendix A. At leading collapsed order, $k_1 = k_2 = k_{12}/2$ and $k_3 = k_4 = k_{34}/2$. Thus left and right dimensionless seed matrices are the same whenever their hard propagator types agree; their different hard scales are carried by the prefactors below. The common dimensional prefactors of the left and right hard subgraphs are

$$\mathcal{N}_L = (igt)^2 \frac{H^2}{4k_1^3 k_2^3} \left(\frac{k_{12}}{2} \right)^{-2i\tilde{\nu}}, \quad \mathcal{N}_R = (igt)^2 \frac{H^2}{4k_3^3 k_4^3} \left(\frac{k_{34}}{2} \right)^{-2i\tilde{\nu}}. \quad (40)$$

The two symmetric soft lines are (23) and (41). Their spinor Kronecker deltas will be inserted separately for each diagram below, *after* both hard subgraphs have been rotated. The common soft factor is already the bubble subgraph $\mathcal{B}_+^{(S)}(k_s)$ of Eq. (29); no new coefficient is needed. If $\mathcal{W}_{+,\ell}$ denotes the dimensionless spinor contraction computed below, the contribution of each surviving diagram before extracting the universal momentum factor is

$$\langle \delta h^4 \rangle'_\ell \Big|_{S,+} = -N_c \mathcal{B}_+^{(S)}(k_s) \mathcal{N}_L \mathcal{N}_R \mathcal{W}_{+,\ell}, \quad \ell \in \mathcal{G}_{AA}. \quad (41)$$

Below, $\hat{\mathcal{Q}}_{+,\alpha\beta}$ and $\check{\mathcal{Q}}_{+,\dot{\alpha}\dot{\beta}}$ denote the seed sums with their original two lower indices. By contrast, $\mathcal{Q}_+^{NN}$ and $\mathcal{Q}_+^{\tilde{N}\tilde{N}}$ denote the mixed-index matrices obtained at the contraction step.

Diagram 1: $NNNN$. Both hard lines are type 2, so their integral seeds are the unraised $\hat{\mathcal{I}}_{ab,\alpha\beta}^{p_1 p_2}$ of Eq. (134). The left seed sum in the z frame is

$$\begin{aligned} \frac{\hat{\mathcal{T}}_{+,1}^{(L)}(\hat{\mathbf{z}})}{\mathcal{N}_L} &= \sum_{a,b=\pm} ab \left[\hat{\mathcal{I}}_{ab}^{p,p} + ia \hat{\mathcal{I}}_{ab}^{p+1,p} + ib \hat{\mathcal{I}}_{ab}^{p,p+1} - ab \hat{\mathcal{I}}_{ab}^{p+1,p+1} \right] (\hat{\mathbf{z}}) \\ &= \hat{\mathcal{Q}}_+(\hat{\mathbf{z}}) = \hat{q}_1 \sigma^1 + i \hat{q}_0 \sigma^2. \end{aligned} \quad (42)$$

The two Pauli matrices here display a matrix with *only* 12 and 21 entries. Its coefficients are read directly from those entries:

$$\hat{q}_0 = \frac{1}{2} (\hat{\mathcal{Q}}_{+,12} - \hat{\mathcal{Q}}_{+,21}), \quad \hat{q}_1 = \frac{1}{2} (\hat{\mathcal{Q}}_{+,12} + \hat{\mathcal{Q}}_{+,21}). \quad (43)$$

The right seed is identical apart from the overall factor: $\hat{\mathcal{T}}_{+,1}^{(R)}(\hat{\mathbf{z}}) = \mathcal{N}_R \hat{\mathcal{Q}}_+(\hat{\mathbf{z}})$. For the rotation, use the spin-1/2 matrix of Ref. [2],

$$U(\theta, \phi) = \begin{pmatrix} e^{-i\phi/2} \cos(\theta/2) & -e^{-i\phi/2} \sin(\theta/2) \\ e^{+i\phi/2} \sin(\theta/2) & e^{+i\phi/2} \cos(\theta/2) \end{pmatrix}. \quad (44)$$

As in the gauge-boson box, take

$$\begin{aligned}
\mathbf{k}_s &= k_s(0, 0, 1), \\
\mathbf{k}_1 &= k_1(\sin \theta_1, 0, \cos \theta_1), \\
\mathbf{k}_3 &= k_3(\sin \theta_3 \cos \phi_3, \sin \theta_3 \sin \phi_3, \cos \theta_3), \\
c_{13} &\equiv \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}_3 = \cos \theta_1 \cos \theta_3 + \sin \theta_1 \sin \theta_3 \cos \phi_3.
\end{aligned} \tag{45}$$

Set $U_1 = U(\theta_1, 0)$ and $U_3 = U(\theta_3, \phi_3)$. Table 1 of Ref. [2] gives the $U\hat{U}U^T$ rule for two lower undotted indices. We therefore rotate the left and right hard tensors separately, with *no* index conversion:

$$\hat{\mathcal{T}}_{+,1}^{(L)}(\hat{\mathbf{k}}_1) = \mathcal{N}_L U_1(\hat{q}_1 \sigma^1 + i\hat{q}_0 \sigma^2) U_1^T, \tag{46}$$

$$\hat{\mathcal{T}}_{+,1}^{(R)}(\hat{\mathbf{k}}_3) = \mathcal{N}_R U_3(\hat{q}_1 \sigma^1 + i\hat{q}_0 \sigma^2) U_3^T. \tag{47}$$

No index has been raised in Eqs. (42)–(47). At the two vertices we now raise the *second* index of each hard tensor, as in Eq. (35), and join their open ends with the two soft deltas. We use $\epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $\hat{\mathcal{Q}}_+(\hat{\mathbf{z}})\epsilon^{-1} = \hat{q}_0 \sigma^0 + \hat{q}_1 \sigma^3$, and $U_i^T \epsilon^{-1} = \epsilon^{-1} U_i^\dagger$. After stripping $\mathcal{N}_L \mathcal{N}_R$, the explicit spinor chain is

$$\begin{aligned}
\mathcal{W}_{+,1} &= [\hat{\mathcal{Q}}_+(\hat{\mathbf{k}}_1)\epsilon^{-1}]^{\alpha_1}_{\alpha_2} \delta^{\alpha_2}_{\alpha_3} [\hat{\mathcal{Q}}_+(\hat{\mathbf{k}}_3)\epsilon^{-1}]^{\alpha_3}_{\alpha_4} \delta^{\alpha_4}_{\alpha_1} \\
&= \text{tr}_2 \left[\hat{\mathcal{Q}}_+(\hat{\mathbf{k}}_1)\epsilon^{-1} \hat{\mathcal{Q}}_+(\hat{\mathbf{k}}_3)\epsilon^{-1} \right] \\
&= \text{tr}_2 \left[U_1(\hat{q}_0 \sigma^0 + \hat{q}_1 \sigma^3) U_1^\dagger U_3(\hat{q}_0 \sigma^0 + \hat{q}_1 \sigma^3) U_3^\dagger \right] \\
&= \text{tr}_2 \left[(\hat{q}_0 \sigma^0 + \hat{q}_1 \hat{\mathbf{k}}_1 \cdot \boldsymbol{\sigma})(\hat{q}_0 \sigma^0 + \hat{q}_1 \hat{\mathbf{k}}_3 \cdot \boldsymbol{\sigma}) \right] \\
&= 2(\hat{q}_0^2 + \hat{q}_1^2 c_{13}).
\end{aligned} \tag{48}$$

Diagram 7: $N\bar{N}\bar{N}N$. The left hard line is type-1 D , whereas the right one is its reversed-flow partner $\bar{D}$. Their axis-frame integral-seed sums are

$$\begin{aligned}
\frac{\mathcal{T}_{+,7}^{(L)}(\hat{\mathbf{z}})}{\mathcal{N}_L} &= \sum_{a,b=\pm} ab \left[\mathcal{I}_{ab}^{N\bar{N};p,p} + i a \mathcal{I}_{ab}^{N\bar{N};p+1,p} + i b \mathcal{I}_{ab}^{N\bar{N};p,p+1} - ab \mathcal{I}_{ab}^{N\bar{N};p+1,p+1} \right] (\hat{\mathbf{z}}) \\
&= \mathcal{Q}_+^{N\bar{N}}(\hat{\mathbf{z}}) = q_0 \sigma^0 + q_1 \sigma^3, \\
\frac{\mathcal{T}_{+,7}^{(R)}(\hat{\mathbf{z}})}{\mathcal{N}_R} &= \sum_{c,d=\pm} cd \left[\mathcal{I}_{cd}^{\bar{N}N;p,p} + i c \mathcal{I}_{cd}^{\bar{N}N;p+1,p} + i d \mathcal{I}_{cd}^{\bar{N}N;p,p+1} - cd \mathcal{I}_{cd}^{\bar{N}N;p+1,p+1} \right] (\hat{\mathbf{z}}) \\
&= \mathcal{Q}_+^{\bar{N}N}(\hat{\mathbf{z}}) = \bar{q}_0 \sigma^0 + \bar{q}_1 \bar{\sigma}^3.
\end{aligned} \tag{49}$$

Here $\bar{\sigma}^i = -\sigma^i$; the four coefficients are simply the mean and half-difference of the *diagonal* entries in these two seed sums. Writing $Q_{jj} = [\mathcal{Q}_+^{N\bar{N}}(\hat{\mathbf{z}})]_{jj}$ and $\bar{Q}_{jj} = [\mathcal{Q}_+^{\bar{N}N}(\hat{\mathbf{z}})]_{jj}$,

$$\begin{aligned}
q_0 &= \frac{1}{2}(Q_{11} + Q_{22}), & q_1 &= \frac{1}{2}(Q_{11} - Q_{22}), \\
\bar{q}_0 &= \frac{1}{2}(\bar{Q}_{11} + \bar{Q}_{22}), & \bar{q}_1 &= \frac{1}{2}(\bar{Q}_{22} - \bar{Q}_{11}).
\end{aligned} \tag{50}$$

Here $\mathcal{I}^{\bar{N}N}$ is constructed from $\mathcal{I}^{N\bar{N}}$ by the reversal identity in Eq. (66); its SK labels c, d refer to vertices 3 and 4. The two type-1 seeds already have one dotted and one undotted index. We

rotate them, without any additional index conversion, by the $UQU^\dagger$ rule of Table 1 of Ref. [2] to the two physical hard directions:

$$\mathcal{T}_{+,7}^{(L)}(\hat{\mathbf{k}}_1) = \mathcal{N}_L U_1(q_0\sigma^0 + q_1\sigma^3)U_1^\dagger, \quad (51)$$

$$\mathcal{T}_{+,7}^{(R)}(\hat{\mathbf{k}}_3) = \mathcal{N}_R U_3(\bar{q}_0\sigma^0 + \bar{q}_1\bar{\sigma}^3)U_3^\dagger. \quad (52)$$

Finally the two soft lines, of types 3 and 2 respectively, each supply the identity in the appropriate spinor space. Inserting these deltas and using $U_i\sigma^3U_i^\dagger = \hat{\mathbf{k}}_i \cdot \sigma$ and $\bar{\sigma}^i = -\sigma^i$ gives

$$\begin{aligned} \mathcal{W}_{+,7} &= [\mathcal{Q}_+^{N\bar{N}}(\hat{\mathbf{k}}_1)]^{\alpha_1}_{\dot{\alpha}_2} \delta^{\dot{\alpha}_2}_{\dot{\alpha}_3} [\mathcal{Q}_+^{\bar{N}N}(\hat{\mathbf{k}}_3)]^{\dot{\alpha}_3}_{\alpha_4} \delta^{\alpha_4}_{\alpha_1} \\ &= \text{tr}_2 \left[(q_0\sigma^0 + q_1\hat{\mathbf{k}}_1 \cdot \sigma)(\bar{q}_0\sigma^0 + \bar{q}_1\hat{\mathbf{k}}_3 \cdot \bar{\sigma}) \right] \\ &= 2(q_0\bar{q}_0 - q_1\bar{q}_1 c_{13}). \end{aligned} \quad (53)$$

Diagram 10: $\bar{N}N\bar{N}\bar{N}$. This exchanges the two type-1 hard seeds of diagram 7. Explicitly, the axis-frame left and right seed sums are the $\bar{N}N$ and $N\bar{N}$ finite sums in Eq. (49), respectively; the left and right prefactors remain $\mathcal{N}_L$ and $\mathcal{N}_R$:

$$\begin{aligned} \mathcal{T}_{+,10}^{(L)}(\hat{\mathbf{z}}) &= \mathcal{N}_L \sum_{a,b=\pm} ab \left[\mathcal{I}_{ab}^{\bar{N}N;p,p} + ia\mathcal{I}_{ab}^{\bar{N}N;p+1,p} + ib\mathcal{I}_{ab}^{\bar{N}N;p,p+1} - ab\mathcal{I}_{ab}^{\bar{N}N;p+1,p+1} \right](\hat{\mathbf{z}}) \\ &= \mathcal{N}_L(\bar{q}_0\sigma^0 + \bar{q}_1\bar{\sigma}^3), \\ \mathcal{T}_{+,10}^{(R)}(\hat{\mathbf{z}}) &= \mathcal{N}_R \sum_{c,d=\pm} cd \left[\mathcal{I}_{cd}^{N\bar{N};p,p} + ic\mathcal{I}_{cd}^{N\bar{N};p+1,p} + id\mathcal{I}_{cd}^{N\bar{N};p,p+1} - cd\mathcal{I}_{cd}^{N\bar{N};p+1,p+1} \right](\hat{\mathbf{z}}) \\ &= \mathcal{N}_R(q_0\sigma^0 + q_1\sigma^3). \end{aligned} \quad (54)$$

The unaltered spinor indices of these two type-1 seeds rotate as

$$\mathcal{T}_{+,10}^{(L)}(\hat{\mathbf{k}}_1) = \mathcal{N}_L U_1(\bar{q}_0\sigma^0 + \bar{q}_1\bar{\sigma}^3)U_1^\dagger, \quad (55)$$

$$\mathcal{T}_{+,10}^{(R)}(\hat{\mathbf{k}}_3) = \mathcal{N}_R U_3(q_0\sigma^0 + q_1\sigma^3)U_3^\dagger. \quad (56)$$

Only after this rotation do we insert the two soft deltas, now of types 2 and 3 in reversed order. Their index chain yields

$$\begin{aligned} \mathcal{W}_{+,10} &= [\mathcal{Q}_+^{\bar{N}N}(\hat{\mathbf{k}}_1)]^{\dot{\alpha}_1}_{\alpha_2} \delta^{\alpha_2}_{\alpha_3} [\mathcal{Q}_+^{N\bar{N}}(\hat{\mathbf{k}}_3)]^{\alpha_3}_{\dot{\alpha}_4} \delta^{\dot{\alpha}_4}_{\dot{\alpha}_1} \\ &= \text{tr}_2 \left[(\bar{q}_0\sigma^0 + \bar{q}_1\hat{\mathbf{k}}_1 \cdot \bar{\sigma})(q_0\sigma^0 + q_1\hat{\mathbf{k}}_3 \cdot \sigma) \right] \\ &= 2(\bar{q}_0q_0 - \bar{q}_1q_1 c_{13}) = \mathcal{W}_{+,7}. \end{aligned} \quad (57)$$

The equality follows from the symmetry of $\hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}_3$ and the same pair of scalar seed coefficients in the two traces; both diagrams must be retained in the sum.

Diagram 16: $\bar{N}\bar{N}\bar{N}\bar{N}$. Both hard lines are type 3. Their seeds are the unraised $\check{\mathcal{I}}_{ab,\dot{\alpha}\dot{\beta}}^{p_1 p_2}$ of Eq. (141). In the axis frame, the left subgraph is

$$\begin{aligned} \frac{\check{\mathcal{T}}_{+,16}^{(L)}(\hat{\mathbf{z}})}{\mathcal{N}_L} &= \sum_{a,b=\pm} ab \left[\check{\mathcal{I}}_{ab}^{p,p} + ia\check{\mathcal{I}}_{ab}^{p+1,p} + ib\check{\mathcal{I}}_{ab}^{p,p+1} - ab\check{\mathcal{I}}_{ab}^{p+1,p+1} \right](\hat{\mathbf{z}}) \\ &= \check{\mathcal{Q}}_+(\hat{\mathbf{z}}) = -\check{q}_1\sigma^1 - i\check{q}_0\sigma^2. \end{aligned} \quad (58)$$

As in diagram 1, only the 12 and 21 entries occur. The signs appropriate to type 3 are fixed by

$$\check{q}_0 = \frac{1}{2}(\check{\mathcal{Q}}_{+,21} - \check{\mathcal{Q}}_{+,12}), \quad \check{q}_1 = -\frac{1}{2}(\check{\mathcal{Q}}_{+,12} + \check{\mathcal{Q}}_{+,21}). \quad (59)$$

The right subgraph has the same seed sum multiplied by $\mathcal{N}_R$: $\check{\mathcal{T}}_{+,16}^{(R)}(\hat{\mathbf{z}}) = \mathcal{N}_R \check{\mathcal{Q}}_+(\hat{\mathbf{z}})$. We rotate both *lower dotted-index* matrices according to Table 1 of Ref. [2], with no index conversion:

$$\check{\mathcal{T}}_{+,16}^{(L)}(\hat{\mathbf{k}}_1) = -\mathcal{N}_L U_1^*(\check{q}_1 \sigma^1 + i\check{q}_0 \sigma^2) U_1^\dagger, \quad (60)$$

$$\check{\mathcal{T}}_{+,16}^{(R)}(\hat{\mathbf{k}}_3) = -\mathcal{N}_R U_3^*(\check{q}_1 \sigma^1 + i\check{q}_0 \sigma^2) U_3^\dagger. \quad (61)$$

Only now do we raise the *first* dotted index of each tensor with ϵ , as in Eq. (35), and contract with the two type-3 soft-line deltas. Here $\epsilon \check{\mathcal{Q}}_+(\hat{\mathbf{z}}) = \check{q}_0 \sigma^0 + \check{q}_1 \bar{\sigma}^3$ and $\epsilon U_i^* = U_i \epsilon$:

$$\begin{aligned} \mathcal{W}_{+,16} &= [\epsilon \check{\mathcal{Q}}_+(\hat{\mathbf{k}}_1)]^{\dot{\alpha}_1 \dot{\alpha}_2} \delta^{\dot{\alpha}_2 \dot{\alpha}_3} [\epsilon \check{\mathcal{Q}}_+(\hat{\mathbf{k}}_3)]^{\dot{\alpha}_3 \dot{\alpha}_4} \delta^{\dot{\alpha}_4 \dot{\alpha}_1} \\ &= \text{tr}_2 \left[\epsilon \check{\mathcal{Q}}_+(\hat{\mathbf{k}}_1) \epsilon \check{\mathcal{Q}}_+(\hat{\mathbf{k}}_3) \right] \\ &= \text{tr}_2 \left[U_1 (\check{q}_0 \sigma^0 + \check{q}_1 \bar{\sigma}^3) U_1^\dagger U_3 (\check{q}_0 \sigma^0 + \check{q}_1 \bar{\sigma}^3) U_3^\dagger \right] \\ &= \text{tr}_2 \left[(\check{q}_0 \sigma^0 + \check{q}_1 \hat{\mathbf{k}}_1 \cdot \bar{\sigma}) (\check{q}_0 \sigma^0 + \check{q}_1 \hat{\mathbf{k}}_3 \cdot \bar{\sigma}) \right] \\ &= 2(\check{q}_0^2 + \check{q}_1^2 c_{13}). \end{aligned} \quad (62)$$

Here the angular term is positive because the two factors $\bar{\sigma}^i = -\sigma^i$ supply two minus signs. These four explicit contractions also show why no separate $\cos \theta_1$, $\cos \theta_3$, or $\cos 2\phi_3$ term can occur in the symmetric answer: after the two soft deltas are extracted, the only directional scalar is $\hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}_3 = c_{13}$.

The ϵ conversions used *only in the contraction step* of diagrams 1 and 16 may be checked component by component. For every SK pair and every seed power, in the convention $\epsilon^{12} = \epsilon^{i\dot{j}} = +1$,

$$\begin{aligned} \begin{pmatrix} 0 & \hat{\mathcal{I}}_{ab,12} \\ \hat{\mathcal{I}}_{ab,21} & 0 \end{pmatrix} \epsilon^{-1} &= \begin{pmatrix} \hat{\mathcal{I}}_{ab,12} & 0 \\ 0 & -\hat{\mathcal{I}}_{ab,21} \end{pmatrix}, \\ \epsilon \begin{pmatrix} 0 & \check{\mathcal{I}}_{ab,i\dot{2}} \\ \check{\mathcal{I}}_{ab,\dot{2}i} & 0 \end{pmatrix} &= \begin{pmatrix} \check{\mathcal{I}}_{ab,\dot{2}i} & 0 \\ 0 & -\check{\mathcal{I}}_{ab,i\dot{2}} \end{pmatrix}. \end{aligned} \quad (63)$$

In particular, the axis-frame hard matrices appearing *inside* the traces, not in the preceding rotation equations, are

$$\begin{aligned} \mathcal{Q}_+^{NN}(\hat{\mathbf{z}}) &= \hat{\mathcal{Q}}_+(\hat{\mathbf{z}}) \epsilon^{-1} = \hat{q}_0 \sigma^0 + \hat{q}_1 \sigma^3, \\ \mathcal{Q}_+^{\bar{N}\bar{N}}(\hat{\mathbf{z}}) &= \epsilon \check{\mathcal{Q}}_+(\hat{\mathbf{z}}) = \check{q}_0 \sigma^0 + \check{q}_1 \bar{\sigma}^3, \quad \epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \end{aligned} \quad (64)$$

These identities also explain why rotating the lower-index seed and then raising one index gives the same answer as raising that index first and applying a mixed-index similarity rotation. The two operations commute because $U^T \epsilon^{-1} = \epsilon^{-1} U^\dagger$ and $\epsilon U^* = U \epsilon$. For diagram 16, an overall type-3 sign would cancel between the two hard factors; Eq. (63) fixes that sign for diagrams with a single type-3 hard line in the antisymmetric sector.

For comparison, collect the four seed sums in the mixed-index notation used elsewhere in this article. Let $p = -1 + i\tilde{\nu}$ and $rs \in \{N\tilde{N}, \tilde{N}N, NN, \tilde{N}\tilde{N}\}$. Then

$$\mathcal{Q}_+^{rs}(\hat{\mathbf{k}}) = \sum_{a,b=\pm} ab \left[\mathcal{I}_{ab}^{rs;p,p} + ia\mathcal{I}_{ab}^{rs;p+1,p} + ib\mathcal{I}_{ab}^{rs;p,p+1} - ab\mathcal{I}_{ab}^{rs;p+1,p+1} \right](\hat{\mathbf{k}}). \quad (65)$$

This is the scalar-Yukawa analogue of the subdiagram construction in Ref. [2]: the four terms $(m, n) = (0, 0), (1, 0), (0, 1), (1, 1)$ come from the two massless Higgs polynomials. The $\widehat{D}, \check{D}$ seeds are converted to mixed indices as specified after Eq. (34). Every axis-frame matrix used above is a finite sum of the You integral seeds, rather than an unevaluated time integral. The type-1 matrix $\mathcal{I}^{N\tilde{N}}$ and the *unraised* type-2 matrix $\hat{\mathcal{I}}$ are displayed explicitly in Eqs. (123) and (134). The other two contain no new integral,

$$\begin{aligned} \mathbf{I}_{ab}^{\bar{D};p_1p_2}(\hat{\mathbf{k}}) &= - \left[\mathbf{I}_{ba}^{D;p_2p_1}(-\hat{\mathbf{k}}) \right]^T, \\ \check{\mathcal{I}}_{ab}^{p_1p_2}(\hat{\mathbf{k}}) &= - \left[\hat{\mathcal{I}}_{-a,-b}^{p_1^*p_2^*}(\hat{\mathbf{k}}) \right]^*. \end{aligned} \quad (66)$$

The second relation concerns the *unraised* type-2 and type-3 seeds. Equation (65) is written in mixed-index notation, so Eq. (63) is applied when using it; in the four diagram-by-diagram derivations above, that conversion was made *after* the raw-seed rotation, at the soft-line contraction step. No extra integration is needed to extract any of the eight coefficients. For type 1, use the two diagonal entries in Eq. (50); for types 2 and 3, use the *12* and *21* entries in Eqs. (43) and (59), respectively, followed by the index conversion in Eq. (64).

To make the special-function dependence explicit, all seed entries needed in Eqs. (50) and (65) reduce to

$$\mathcal{R}^{p_1p_2}(x, y) = e^{\frac{i\pi}{2}(p_1-p_2)} \frac{\prod_{j=1}^2 [\Gamma(1+p_j-i\tilde{\nu})\Gamma(1+p_j+i\tilde{\nu})]}{2^{2+p_1+p_2}\Gamma(p_1+x)\Gamma(p_2+y)}, \quad (67)$$

$$\begin{aligned} \mathfrak{J}^{p_1p_2}(x, y, z) &= i 2^{-2-p_1-p_2} e^{-\frac{i\pi}{2}(p_1+p_2)} \pi \operatorname{csch}(2\pi\tilde{\nu}) \Gamma(2+p_1+p_2) \\ &\times \left[\frac{e^{-\pi\tilde{\nu}} \Gamma(1+p_1-i\tilde{\nu}) \Gamma(2+p_1+p_2-2i\tilde{\nu})}{\Gamma(x)} \right. \\ &\quad \times {}_4\tilde{F}_3 \left(\begin{matrix} 2+p_1+p_2, 1+p_1-i\tilde{\nu}, y, 2+p_1+p_2-2i\tilde{\nu} \\ 2+p_1-i\tilde{\nu}, 3+p_1+p_2-z, 1-2i\tilde{\nu} \end{matrix}; -1 \right) - (\tilde{\nu} \rightarrow -\tilde{\nu}) \Big], \end{aligned} \quad (68)$$

$$\begin{aligned} \mathfrak{J}^{p_1p_2}(x, y, z) &= i 2^{-2-p_1-p_2} e^{\frac{i\pi}{2}(p_1+p_2)} \pi \operatorname{csch}(2\pi\tilde{\nu}) \Gamma(2+p_1+p_2) \\ &\times \left[\frac{e^{\pi\tilde{\nu}} \Gamma(1+p_1-i\tilde{\nu}) \Gamma(2+p_1+p_2-2i\tilde{\nu})}{\Gamma(x)} \right. \\ &\quad \times {}_4\tilde{F}_3 \left(\begin{matrix} 2+p_1+p_2, 1+p_1-i\tilde{\nu}, y, 2+p_1+p_2-2i\tilde{\nu} \\ 2+p_1-i\tilde{\nu}, 3+p_1+p_2-z, 1-2i\tilde{\nu} \end{matrix}; -1 \right) - (\tilde{\nu} \rightarrow -\tilde{\nu}) \Big]. \end{aligned} \quad (69)$$

Here ${}_4\tilde{F}_3 = {}_4F_3 / \prod_{j=1}^3 \Gamma(b_j)$ is the regularized hypergeometric function. The finite list of arguments

entering the type-1 block is

$$\begin{aligned}
A_1 &= \tilde{\nu}^2 \mathfrak{I}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, i\tilde{\nu}), & A_2 &= \mathfrak{I}(i\tilde{\nu}, -i\tilde{\nu}, 1 + i\tilde{\nu}), \\
B_1 &= \mathfrak{I}^{p_2 p_1}(i\tilde{\nu}, -i\tilde{\nu}, 1 + i\tilde{\nu}), & B_2 &= \tilde{\nu}^2 \mathfrak{I}^{p_2 p_1}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, i\tilde{\nu}), \\
U_1 &= \tilde{\nu}^2 \mathfrak{K}(2, 2), & U_2 &= \mathfrak{K}(1, 1), \\
V_1 &= e^{-i\pi(p_1 - p_2)} \mathfrak{K}(1, 1), & V_2 &= \tilde{\nu}^2 e^{-i\pi(p_1 - p_2)} \mathfrak{K}(2, 2),
\end{aligned} \tag{70}$$

where the superscript $p_1 p_2$ is understood on every $\mathfrak{I}$ and $\mathfrak{K}$ unless it is shown explicitly. For the type-2 block the corresponding list is

$$\begin{aligned}
S_1 &= \tilde{\nu} \mathfrak{I}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, 1 + i\tilde{\nu}), & S_2 &= \tilde{\nu} \mathfrak{I}(i\tilde{\nu}, -i\tilde{\nu}, i\tilde{\nu}), \\
S_3 &= \tilde{\nu} \mathfrak{I}^{p_2 p_1}(i\tilde{\nu}, -i\tilde{\nu}, i\tilde{\nu}), & S_4 &= \tilde{\nu} \mathfrak{I}^{p_2 p_1}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, 1 + i\tilde{\nu}), \\
W_1 &= \tilde{\nu} \mathfrak{K}(2, 1), & W_2 &= \tilde{\nu} \mathfrak{K}(1, 2).
\end{aligned} \tag{71}$$

Equations (123) and (134) specify where these entries occur in each SK matrix. Hence Eqs. (67)–(71) expose the complete mass dependence; no hard time integral is left unevaluated in the symmetric result.

Summing Eqs. (48), (53), (57), and (62), it is useful to define

$$\begin{aligned}
\mathcal{R}_0(\tilde{\nu}) &= \hat{q}_0^2 + \check{q}_0^2 + 2q_0 \bar{q}_0, \\
\mathcal{R}_1(\tilde{\nu}) &= \hat{q}_1^2 + \check{q}_1^2 - 2q_1 \bar{q}_1.
\end{aligned} \tag{72}$$

The remaining bubble functions can also be written entirely in Gamma functions,

$$\begin{aligned}
\kappa_+ &= i \left[\frac{\Gamma(-2i\tilde{\nu})}{\Gamma(-i\tilde{\nu})} \right]^2, \\
f_0(-i\tilde{\nu}, -i\tilde{\nu}) &= \frac{\Gamma(-\frac{3}{2} - 2i\tilde{\nu}) \Gamma^2(\frac{3}{2} + i\tilde{\nu})}{\Gamma(3 + 2i\tilde{\nu}) \Gamma^2(-i\tilde{\nu})}.
\end{aligned} \tag{73}$$

To extract the gauge-boson-box momentum convention from Eq. (41), combine the already-defined bubble with the two hard prefactors:

$$\mathcal{B}_+^{(S)}(k_s) \mathcal{N}_L \mathcal{N}_R = \frac{g_t^4 H^4 k_s^3}{16 \prod_{j=1}^4 k_j^3} \left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2i\tilde{\nu}} \frac{2^{12i\tilde{\nu}} \kappa_+^2}{(4\pi)^{3/2}} f_0(-i\tilde{\nu}, -i\tilde{\nu}). \tag{74}$$

The factor $2^{12i\tilde{\nu}}$ belongs to this extraction of the common momentum power; it is not a second bubble coefficient. The complete symmetric amplitude for one cyclic ordering is therefore

$$C_{(S)}^{\text{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) = A_0^{(S)}(\tilde{\nu}) + A_{13}^{(S)}(\tilde{\nu}) c_{13}, \tag{75}$$

$$\begin{aligned}
A_0^{(S)}(\tilde{\nu}) &= \frac{2^{1+12i\tilde{\nu}} \kappa_+^2}{(4\pi)^{3/2}} f_0(-i\tilde{\nu}, -i\tilde{\nu}) \mathcal{R}_0(\tilde{\nu}), \\
A_{13}^{(S)}(\tilde{\nu}) &= \frac{2^{1+12i\tilde{\nu}} \kappa_+^2}{(4\pi)^{3/2}} f_0(-i\tilde{\nu}, -i\tilde{\nu}) \mathcal{R}_1(\tilde{\nu}).
\end{aligned} \tag{76}$$

Using the gauge-boson-box normalization, the corresponding four-point function is

$$\langle \delta h^4 \rangle'_{\text{Box}}|_S = -N_c g_t^4 \frac{H^4 k_s^3}{16 k_1^3 k_2^3 k_3^3 k_4^3} \left(\frac{k_s^2}{4 k_{12} k_{34}} \right)^{2i\tilde{\nu}} [A_0^{(S)} + A_{13}^{(S)} c_{13}] + \text{c.c.} + \mathcal{O}\left(\frac{k_s}{k_i}\right). \quad (77)$$

This is the fermionic counterpart of the gauge-boson box result: a mass-dependent amplitude multiplying a complex power of the collapsed momentum ratio. In contrast to the vector case, the soft spinor deltas leave only the monopole and the term linear in c_{13} for a fixed cyclic ordering. The phase $2^{12i\tilde{\nu}}$ combines the normalization of the two soft propagators, the choice of hard scales $k_{12}/2$ and $k_{34}/2$, and the factor 4 in the clock argument; it is convention dependent, whereas the full expression including its complex conjugate is not.

3 Antisymmetric soft-line contribution

We next retain the antisymmetric part that was left for future work in Ref. [2]. At zero chemical potential, Eq. (2.39) and Appendix C.3 of that reference give

$$\rho_f(\tilde{\nu}) \equiv \left[\frac{\Gamma(-2if\tilde{\nu})}{\Gamma(-if\tilde{\nu})} \right]^2, \quad \kappa_f = if\rho_f, \quad (78)$$

$$[D_{ab}(k; \tau, \tau')]_{\text{NLoc},f} = -\rho_f(4k^2\tau\tau')^{if\tilde{\nu}} \Xi_{ab}(\tau, \tau') \hat{\mathbf{k}} \cdot \sigma, \quad (79)$$

$$\Xi_{ab}(\tau, \tau') = \frac{a+b}{2} \text{sgn}(\tau - \tau') + \frac{b-a}{2}, \quad a, b = \pm 1. \quad (80)$$

Thus $\Xi_{-+} = 1$, $\Xi_{+-} = -1$, $\Xi_{++} = \text{sgn}(\tau - \tau')$, and $\Xi_{--} = -\text{sgn}(\tau - \tau')$. For the reversed type-1 line it is convenient to keep the same coefficient $-\rho_f$ and use $\hat{\mathbf{k}} \cdot \bar{\sigma} = -\hat{\mathbf{k}} \cdot \sigma$. The other two propagators obey

$$[\widehat{D}^{(A)}]_{\text{NLoc}} = 0, \quad [\check{D}^{(A)}]_{\text{NLoc}} = 0 \quad (\tilde{\lambda} = 0). \quad (81)$$

Consequently the antisymmetric contribution of rows 1, 7, 10, 16 is identically zero. The remaining twelve rows contain one or two type-1 soft lines and must be retained.

3.1 Loop momentum integral and tensor bubble subgraphs

We follow the gauge-boson box calculation and first do the soft loop momentum. Put $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$ and $x = -if\tilde{\nu}$. Before this integral, the universal nonanalytic factor is

$$\mathcal{L}\mathcal{O}\mathcal{O}\mathcal{P}_{f,\ell}^{\text{soft}} = \left[\prod_{j=1}^4 (-\tau_j)^{if\tilde{\nu}} \right] 2^{4if\tilde{\nu}} \int \frac{d^3\mathbf{q}}{(2\pi)^3} q^{-2x} K^{-2x} \mathcal{R}_{23} \otimes \mathcal{R}_{41}, \quad (82)$$

where $\mathcal{R} = \sigma^0$ for a type-2/type-3 line and is $\hat{\mathbf{q}} \cdot \sigma$ or $\hat{\mathbf{K}} \cdot \sigma$ for a type-1 line. A reversed type-1 line carries $\bar{\sigma} = -\sigma$; this sign will be denoted by $\zeta = -1$. Only scalar, vector, and rank-two tensor moments occur. Using the same loop seeds as the gauge-boson calculation, define

$$f_c(a, b) = \frac{1}{2} f_0\left(a + \frac{1}{2}, b - 1\right) - \frac{1}{2} f_0\left(a - \frac{1}{2}, b\right) - \frac{1}{2} f_0\left(a + \frac{1}{2}, b\right), \quad (83)$$

$$\begin{aligned} f_{cc}(a, b) &= \frac{1}{4} f_0(a + 1, b - 2) + \frac{1}{4} f_0(a - 1, b) + \frac{1}{4} f_0(a + 1, b) \\ &\quad - \frac{1}{2} f_0(a, b - 1) - \frac{1}{2} f_0(a + 1, b - 1) + \frac{1}{2} f_0(a, b). \end{aligned} \quad (84)$$

and

$$\mathcal{V}_q(x) = f_c(x, x), \quad \mathcal{V}_K(x) = -f_c(x, x), \quad (85)$$

$$\mathcal{J}_0(x) = f_0\left(x - \frac{1}{2}, x + \frac{1}{2}\right) - \frac{1}{2}f_0\left(x + \frac{1}{2}, x + \frac{1}{2}\right), \quad (86)$$

$$\mathcal{J}_L(x) = f_{cc}\left(x - \frac{1}{2}, x + \frac{1}{2}\right) + f_c\left(x, x + \frac{1}{2}\right), \quad \mathcal{J}_T(x) = \frac{\mathcal{J}_0(x) - \mathcal{J}_L(x)}{2}. \quad (87)$$

After extracting $k_s^{3+4if\tilde{\nu}}/(4\pi)^{3/2}$, the required moments are

$$\begin{aligned} \int \hat{q}_i &= \mathcal{V}_q \hat{k}_{si}, & \int \hat{K}_i &= \mathcal{V}_K \hat{k}_{si}, \\ \int \hat{K}_i \hat{q}_j &= \mathcal{J}_T(\delta_{ij} - \hat{k}_{si} \hat{k}_{sj}) + \mathcal{J}_L \hat{k}_{si} \hat{k}_{sj}. \end{aligned} \quad (88)$$

The integral signs in Eq. (88) include the common weights $q^{-2x} K^{-2x}$. The tensor trace gives the useful check $2\mathcal{J}_T + \mathcal{J}_L = \mathcal{J}_0$. Restoring the factors stripped above,

$$\mathcal{LOOP}_{f,\ell}^{soft} = \left[\prod_{j=1}^4 (-\tau_j)^{if\tilde{\nu}} \right] \frac{2^{4if\tilde{\nu}} k_s^{3+4if\tilde{\nu}}}{(4\pi)^{3/2}} \mathfrak{B}_{f,\ell} \times (\text{SK/time kernels}), \quad (89)$$

where $\mathfrak{B}_{f,\ell}$ is the vector or tensor in Eq. (88). Thus the soft loop momentum is completely integrated before treating its time-ordering kernel.

3.2 The sixteen rows in the antisymmetric sector

The following table supplies the information needed for every row. The pair in the third column is (X_L, X_R) for the hard edges (12) and (34). A symbol $D_A(K)$ or $\bar{D}_A(q)$ denotes a type-1 antisymmetric soft line and fixes both its momentum and its sign ζ , with $\zeta_D = +1$ and $\zeta_{\bar{D}} = -1$.

ℓ	word	(X_L, X_R)	edge (23)	edge (41)
1	NNNN	$(\hat{D}, \hat{D})$	$\hat{D}_S$	$\hat{D}_S$
2	NNN $\bar{N}$	$(\hat{D}, D)$	$\hat{D}_S$	$\bar{D}_A(q)$
3	NN $\bar{N}$ N	$(\hat{D}, \bar{D})$	$D_A(K)$	$\hat{D}_S$
4	NN $\bar{N}$ $\bar{N}$	$(\hat{D}, \check{D})$	$D_A(K)$	$\bar{D}_A(q)$
5	N $\bar{N}$ NN	$(D, \hat{D})$	$\bar{D}_A(K)$	$\hat{D}_S$
6	N $\bar{N}$ N $\bar{N}$	(D, D)	$\bar{D}_A(K)$	$\bar{D}_A(q)$
7	N $\bar{N}$ $\bar{N}$ N	$(D, \bar{D})$	$\check{D}_S$	$\hat{D}_S$
8	N $\bar{N}$ $\bar{N}$ $\bar{N}$	$(D, \check{D})$	$\check{D}_S$	$\bar{D}_A(q)$
9	$\bar{N}$ NNN	$(\bar{D}, \hat{D})$	$\hat{D}_S$	$D_A(q)$
10	$\bar{N}$ NN $\bar{N}$	$(\bar{D}, D)$	$\hat{D}_S$	$\check{D}_S$
11	$\bar{N}$ N $\bar{N}$ N	$(\bar{D}, \bar{D})$	$D_A(K)$	$D_A(q)$
12	$\bar{N}$ N $\bar{N}$ $\bar{N}$	$(\bar{D}, \check{D})$	$D_A(K)$	$\check{D}_S$
13	$\bar{N}$ $\bar{N}$ NN	$(\check{D}, \hat{D})$	$\bar{D}_A(K)$	$D_A(q)$
14	$\bar{N}$ $\bar{N}$ N $\bar{N}$	$(\check{D}, D)$	$\bar{D}_A(K)$	$\check{D}_S$
15	$\bar{N}$ $\bar{N}$ $\bar{N}$ N	$(\check{D}, \bar{D})$	$\check{D}_S$	$D_A(q)$
16	$\bar{N}$ $\bar{N}$ $\bar{N}$ $\bar{N}$	$(\check{D}, \check{D})$	$\check{D}_S$	$\check{D}_S$

Rows 1, 7, 10, 16 contain no antisymmetric line and hence contribute zero to the present sector. The remaining rows fall into

$$\mathcal{G}_{AN} = \{2, 8, 9, 15\}, \quad \mathcal{G}_{NA} = \{3, 5, 12, 14\}, \quad \mathcal{G}_{NN} = \{4, 6, 11, 13\}, \quad (90)$$

where AN has a type-1 line on (41), NA has one on (23), and NN has one on each soft edge. This accounts for all $4 + 4 + 4 + 4 = 16$ rows.

3.3 Time kernels and energy-deformed You seeds

The sign kernel is separated by the exact spectral identity

$$\text{sgn}(\tau - \tau') = \frac{1}{i\pi} \text{PV} \int_{-\infty}^{+\infty} \frac{d\omega}{\omega} e^{i\omega(\tau - \tau')}. \quad (91)$$

For each fixed pair of SK labels introduce

$$C_{uv}^{(0)} = \frac{v - u}{2}, \quad C_{uv}^{(1)} = \frac{u + v}{2}, \quad \Xi_{uv} = C_{uv}^{(0)} + C_{uv}^{(1)} \text{sgn}(\tau - \tau'). \quad (92)$$

Thus an opposite-SK type-1 line selects the algebraic term $C^{(0)}$, while an equal-SK line selects one Hilbert transform through $C^{(1)}$. This observation avoids any double counting of SK sums.

The hard blocks must therefore be kept branch resolved. Define the energy-deformed Yukawa seed, without an SK sum, by

$$\mathbf{Q}_{f;ab}^X(\hat{\mathbf{k}}; u, v) \equiv \int_0^\infty dz_1 dz_2 z_1^{p_f} z_2^{p_f} (1 + ia z_1)(1 + ib z_2) e^{-iaz_1 - ibz_2 + iuz_1 + ivz_2} \mathbf{D}_{ab}^X(\hat{\mathbf{k}}; z_1, z_2), \quad (93)$$

where $p_f = -1 + i f \tilde{\nu}$. It is important that Eq. (93) contains no factor ab ; the four branch labels are summed only after the cross-edge kernels have been inserted. For $X = \widehat{D}, \check{D}$, both $\mathbf{D}^X$ here and the corresponding $\mathbf{I}^X$ below are the mixed-index matrices obtained from Eq. (35); in particular the raw type-3 seed is multiplied by ϵ on the *left*, as in Eq. (63). The dependence on the special functions of Ref. [2] is made explicit by the convergent, $i\epsilon$ -regulated expansion

$$\begin{aligned} \mathbf{Q}_{f;ab}^X(u, v) = \sum_{r,s=0}^{\infty} \frac{(iu)^r (iv)^s}{r!s!} & \left[\mathbf{I}_{ab}^{X;p_f+r,p_f+s} + ia \mathbf{I}_{ab}^{X;p_f+r+1,p_f+s} \right. \\ & \left. + ib \mathbf{I}_{ab}^{X;p_f+r,p_f+s+1} - ab \mathbf{I}_{ab}^{X;p_f+r+1,p_f+s+1} \right]. \end{aligned} \quad (94)$$

Every matrix $\mathbf{I}_{ab}^{X;p_1 p_2}$ on the right-hand side is a You seed from Appendix A, with the anomalous index conversion of Eq. (63) applied where needed; hence every term is an explicit combination of Gamma functions and ${}_4\tilde{F}_3(-1)$. Equation (94), analytically continued from the regulated half-plane, is also a useful numerical representation of the Hilbert transforms below. At $u = v = 0$ it reduces to the finite four-seed combination in Eq. (65) before the SK sum.

Let $\epsilon_{23}, \epsilon_{41} \in \{0, 1\}$ indicate whether the algebraic or sign term in Eq. (92) is selected. Define

$$\mathbb{P}_{\epsilon_{23}\epsilon_{41}} F \equiv \prod_{e \in \{23, 41\}: \epsilon_e=1} \left[\frac{1}{i\pi} \text{PV} \int_{-\infty}^{+\infty} \frac{d\Omega_e}{\Omega_e} \right] F \left(\frac{2\epsilon_{41}\Omega_{41}}{k_{12}}, -\frac{2\epsilon_{23}\Omega_{23}}{k_{12}}, \frac{2\epsilon_{23}\Omega_{23}}{k_{34}}, -\frac{2\epsilon_{41}\Omega_{41}}{k_{34}} \right), \quad (95)$$

where an absent Ω_e is set to zero. The four arguments of F are respectively $(u_L, v_L; u_R, v_R)$. They follow directly from $e^{i\Omega_{23}(\tau_2-\tau_3)}$ and $e^{i\Omega_{41}(\tau_4-\tau_1)}$.

For row ℓ , decompose the two branch-resolved hard blocks as

$$\begin{aligned}\mathbf{Q}_{+,ab}^{X_L}(\hat{\mathbf{k}}_1; u_L, v_L) &= L_{0,ab}^\ell \sigma^0 + L_{1,ab}^\ell \hat{\mathbf{k}}_1 \cdot \sigma, \\ \mathbf{Q}_{+,cd}^{X_R}(\hat{\mathbf{k}}_3; u_R, v_R) &= R_{0,cd}^\ell \sigma^0 + R_{1,cd}^\ell \hat{\mathbf{k}}_3 \cdot \sigma.\end{aligned}\tag{96}$$

As in Eq. (50), the sign appropriate to a dotted block has already been absorbed into L_1 or R_1 . For $m, n = 0, 1$, define the exact hard functional

$$\mathcal{H}_\ell^{mn} \equiv \sum_{a,b,c,d=\pm} abcd \sum_{\epsilon_{23}, \epsilon_{41}} C_{bc}^{(\epsilon_{23})} C_{da}^{(\epsilon_{41})}, \mathbb{P}_{\epsilon_{23}\epsilon_{41}} [L_{m,ab}^\ell R_{n,cd}^\ell]. \tag{97}$$

For a type-2/type-3 soft edge, the corresponding ϵ sum in Eq. (97) is replaced by the single value $\epsilon = 0$ with $C^{(0)} = 1$. For a type-1 edge both values are retained. Thus $\mathcal{H}_\ell^{mn}$ contains a finite piece and at most one PV integral for AN, NA , and at most two PV integrals for NN . Equations (94), (95), and (97) are the corrected branch-resolved form of the antisymmetric master formula. To write it uniformly, set

$$\chi_e^\ell = \begin{cases} \Xi_e, & \text{if edge } e \text{ is type-1 in row } \ell, \\ 1, & \text{if edge } e \text{ is type-2/type-3 in row } \ell. \end{cases} \tag{98}$$

Here Σ_e is the raised soft spinor matrix: it is σ^0 on a type-2/type-3 edge, and $\zeta_e \hat{\mathbf{p}}_e \cdot \sigma$ on a type-1 edge, with $\mathbf{p}_{23} = \mathbf{K}$ and $\mathbf{p}_{41} = \mathbf{q}$. The bracket $[\cdots]_{\text{spectral}}$ means that the vector or tensor bubble moment in Eq. (88) and the Hilbert operators in Eq. (95) are applied before the branch sum. Then

$$\mathcal{W}_{+,\ell}^{(A)} = \sum_{a,b,c,d=\pm} abcd \left[\chi_{23,bc}^\ell \chi_{41,da}^\ell \text{tr}_2(\mathbf{Q}_{+,ab}^{X_L} \Sigma_{23} \mathbf{Q}_{+,cd}^{X_R} \Sigma_{41}) \right]_{\text{spectral}}. \tag{99}$$

3.4 Spinor contractions for the twelve nonzero rows

Write

$$\begin{aligned}c_1 &= \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}_s = \cos \theta_1, & c_3 &= \hat{\mathbf{k}}_3 \cdot \hat{\mathbf{k}}_s = \cos \theta_3, \\ p_{13} &= \hat{\mathbf{k}}_s \cdot (\hat{\mathbf{k}}_1 \times \hat{\mathbf{k}}_3) = \sin \theta_1 \sin \theta_3 \sin \phi_3.\end{aligned}\tag{100}$$

For $\ell \in \mathcal{G}_{AN}$, the type-1 line is (41) and the Pauli matrix lies to the right of the right hard block. Its contraction is

$$\mathcal{W}_\ell^{AN} = 2\zeta_{41}^\ell \mathcal{V}_q \left(\mathcal{H}_\ell^{01} c_3 + \mathcal{H}_\ell^{10} c_1 + i\mathcal{H}_\ell^{11} p_{13} \right), \quad \ell = 2, 8, 9, 15. \tag{101}$$

For $\ell \in \mathcal{G}_{NA}$, the type-1 matrix is between the two hard blocks, giving the opposite sign for the three-Pauli trace,

$$\mathcal{W}_\ell^{NA} = 2\zeta_{23}^\ell \mathcal{V}_K \left(\mathcal{H}_\ell^{01} c_3 + \mathcal{H}_\ell^{10} c_1 - i\mathcal{H}_\ell^{11} p_{13} \right), \quad \ell = 3, 5, 12, 14. \tag{102}$$

Finally, contracting the symmetric tensor in Eq. (88) with $\text{tr}(\mathbf{Q}_L \sigma^i \mathbf{Q}_R \sigma^j)$ gives

$$\mathcal{W}_\ell^{NN} = 2\zeta_{23}^\ell \zeta_{41}^\ell \left\{ \mathcal{J}_0 \mathcal{H}_\ell^{00} + \mathcal{H}_\ell^{11} \left[-\mathcal{J}_L c_{13} + 2(\mathcal{J}_L - \mathcal{J}_T) c_1 c_3 \right] \right\}, \quad \ell = 4, 6, 11, 13. \quad (103)$$

All bubble functions in Eqs. (101)–(103) are evaluated at $x = -i\tilde{\nu}$. The signs read directly from the table:

$$\begin{aligned} (\zeta_{41}^2, \zeta_{41}^8, \zeta_{41}^9, \zeta_{41}^{15}) &= (-1, -1, +1, +1), \\ (\zeta_{23}^3, \zeta_{23}^5, \zeta_{23}^{12}, \zeta_{23}^{14}) &= (+1, -1, +1, -1), \\ (\zeta_{23} \zeta_{41})_{4,6,11,13} &= (-1, +1, +1, -1). \end{aligned} \quad (104)$$

Equations (101)–(104), together with the hard-pair column of the table, are the individual result for each of the twelve nonzero rows. No spinor index or loop momentum remains uncontracted.

For clarity, the twelve answers can be displayed without any class notation. Define

$$\begin{aligned} F_\pm^\ell &= \mathcal{H}_\ell^{01} c_3 + \mathcal{H}_\ell^{10} c_1 \pm i \mathcal{H}_\ell^{11} p_{13}, \\ G^\ell &= \mathcal{J}_0 \mathcal{H}_\ell^{00} + \mathcal{H}_\ell^{11} \left[-\mathcal{J}_L c_{13} + 2(\mathcal{J}_L - \mathcal{J}_T) c_1 c_3 \right]. \end{aligned} \quad (105)$$

The superscript ℓ fixes the hard pair listed in the sixteen-row table, so that, for example, $\mathcal{H}_2^{mn} = \mathcal{H}^{mn}[\widehat{D}, D]$ and $\mathcal{H}_{13}^{mn} = \mathcal{H}^{mn}[\check{D}, \widehat{D}]$. The nonzero contractions are

$$\begin{aligned} \mathcal{W}_2^{AN} &= -2\mathcal{V}_q F_+^2, & \mathcal{W}_8^{AN} &= -2\mathcal{V}_q F_+^8, & \mathcal{W}_9^{AN} &= +2\mathcal{V}_q F_+^9, & \mathcal{W}_{15}^{AN} &= +2\mathcal{V}_q F_+^{15}, \\ \mathcal{W}_3^{NA} &= +2\mathcal{V}_K F_-^3, & \mathcal{W}_5^{NA} &= -2\mathcal{V}_K F_-^5, & \mathcal{W}_{12}^{NA} &= +2\mathcal{V}_K F_-^{12}, & \mathcal{W}_{14}^{NA} &= -2\mathcal{V}_K F_-^{14}, \\ \mathcal{W}_4^{NN} &= -2G^4, & \mathcal{W}_6^{NN} &= +2G^6, & \mathcal{W}_{11}^{NN} &= +2G^{11}, & \mathcal{W}_{13}^{NN} &= -2G^{13}. \end{aligned} \quad (106)$$

Together with $\mathcal{W}_{1,7,10,16}^{(A)} = 0$, Eq. (106) is an explicit sixteen-row audit of the antisymmetric sector.

3.5 Nonlocal signal and angular coefficients

One symmetric type-2/type-3 line contributes $\kappa_+ = i\rho_+$ and one antisymmetric type-1 line contributes $-\rho_+$. Two type-1 lines contribute ρ_+^2 . It follows that

$$C_{(A)}^{\text{Box}} = \mathcal{P}_A(\tilde{\nu}) \left[-i \sum_{\ell \in \mathcal{G}_{AN}} \mathcal{W}_\ell^{AN} - i \sum_{\ell \in \mathcal{G}_{NA}} \mathcal{W}_\ell^{NA} + \sum_{\ell \in \mathcal{G}_{NN}} \mathcal{W}_\ell^{NN} \right], \quad \mathcal{P}_A(\tilde{\nu}) = \frac{2^{12i\tilde{\nu}} \rho_+^2}{(4\pi)^{3/2}}. \quad (107)$$

This already has a form directly suitable for numerical evaluation. To display the angle dependence as in the gauge-boson box, write

$$C_{(A)}^{\text{Box}} = A_0^{(A)} + A_1^{(A)} c_1 + A_3^{(A)} c_3 + A_{13}^{(A)} c_{13} + A_L^{(A)} c_1 c_3 + A_P^{(A)} p_{13}. \quad (108)$$

The six mass- and ratio-dependent coefficients are

$$A_1^{(A)} = -2i\mathcal{P}_A \left[\sum_{\ell \in \mathcal{G}_{AN}} \zeta_{41}^\ell \mathcal{V}_q \mathcal{H}_\ell^{10} + \sum_{\ell \in \mathcal{G}_{NA}} \zeta_{23}^\ell \mathcal{V}_K \mathcal{H}_\ell^{10} \right], \quad (109a)$$

$$A_3^{(A)} = -2i\mathcal{P}_A \left[\sum_{\ell \in \mathcal{G}_{AN}} \zeta_{41}^\ell \mathcal{V}_q \mathcal{H}_\ell^{01} + \sum_{\ell \in \mathcal{G}_{NA}} \zeta_{23}^\ell \mathcal{V}_K \mathcal{H}_\ell^{01} \right], \quad (109b)$$

$$A_0^{(A)} = 2\mathcal{P}_A \sum_{\ell \in \mathcal{G}_{NN}} \zeta_{23}^\ell \zeta_{41}^\ell \mathcal{J}_0 \mathcal{H}_\ell^{00}, \quad (109c)$$

$$A_{13}^{(A)} = -2\mathcal{P}_A \sum_{\ell \in \mathcal{G}_{NN}} \zeta_{23}^\ell \zeta_{41}^\ell \mathcal{J}_L \mathcal{H}_\ell^{11}, \quad (109d)$$

$$A_L^{(A)} = 4\mathcal{P}_A \sum_{\ell \in \mathcal{G}_{NN}} \zeta_{23}^\ell \zeta_{41}^\ell (\mathcal{J}_L - \mathcal{J}_T) \mathcal{H}_\ell^{11}, \quad (109e)$$

$$A_P^{(A)} = 2\mathcal{P}_A \left[\sum_{\ell \in \mathcal{G}_{AN}} \zeta_{41}^\ell \mathcal{V}_q \mathcal{H}_\ell^{11} - \sum_{\ell \in \mathcal{G}_{NA}} \zeta_{23}^\ell \mathcal{V}_K \mathcal{H}_\ell^{11} \right]. \quad (109f)$$

Here a sum labelled by a class runs over the four rows in Eq. (90). The coefficients depend on $\tilde{\nu}$ through ρ_+ , the Gamma-function bubble seeds, and the Gamma- $_{-4}\tilde{F}_3$ series in Eq. (94). They also depend on the hard-scale ratio $r = k_{12}/k_{34}$ through the frequency arguments in Eq. (95). Thus their mass and kinematic dependence is explicit.

Charge conjugation pairs the rows as (2, 15), (8, 9), (3, 14), (5, 12), (4, 13), and (6, 11). Applying Eq. (66) inside Eq. (97), followed where necessary by the harmless PV change of variables $\Omega_e \rightarrow -\Omega_e$, shows that their parity-odd pieces cancel pairwise, so

$$A_P^{(A)} = 0. \quad (110)$$

The exact antisymmetric answer for one fixed cyclic ordering is therefore the five-term polynomial in Eq. (108) with its last term omitted. This is the most direct analogue of the gauge-boson angular decomposition.

The complete answer for a fixed cyclic ordering is

$$\langle \delta h^4 \rangle'_{Box} = -N_c g_t^4 \frac{H^4 k_s^3}{16 k_1^3 k_2^3 k_3^3 k_4^3} \left(\frac{k_s^2}{4 k_{12} k_{34}} \right)^{2i\tilde{\nu}} C^{Box}(\tilde{\nu}, r, \theta_1, \theta_3, \phi_3) + \text{c.c.}, \quad (111)$$

$$\begin{aligned} C^{Box} &= C_{(S)}^{Box} + C_{(A)}^{Box} \\ &= A_0 + A_1 c_1 + A_3 c_3 + A_{13} c_{13} + A_L c_{13}, \end{aligned} \quad (112)$$

where

$$\begin{aligned} A_0 &= A_0^{(S)} + A_0^{(A)}, & A_{13} &= A_{13}^{(S)} + A_{13}^{(A)}, \\ A_1 &= A_1^{(A)}, & A_3 &= A_3^{(A)}, & A_L &= A_L^{(A)}. \end{aligned} \quad (113)$$

The complex power is identical to the symmetric and gauge-boson boxes. The only new objects are one- and two-dimensional Hilbert transforms of known fermion seeds. A finite Gamma-function expression for these transforms is not available in Ref. [2]; that paper explicitly identifies the

antisymmetric time ordering as the obstruction to direct cutting. Equations (94) and (95) isolate this obstruction without leaving any four-time integral.

For later reference, the symmetric angular result is

$$C_{(S)}^{\text{Box}} = A_0^{(S)} + A_{13}^{(S)} c_{13}. \quad (114)$$

Combining it with Eq. (110), the angular dependence of one cyclic box is

$$C_{\text{cyclic}}^{\text{Box}} = A_0 + A_{13} c_{13} + A_1 c_1 + A_3 c_3 + A_L c_1 c_3. \quad (115)$$

There is no $\cos 2\phi_3$ term.

The observable correlator of identical Higgs fluctuations also includes crossed cyclic orderings. At leading order, exchanging the two legs in the left pair sends $\hat{\mathbf{k}}_1 \rightarrow -\hat{\mathbf{k}}_1$, and the analogous exchange on the right sends $\hat{\mathbf{k}}_3 \rightarrow -\hat{\mathbf{k}}_3$. Averaging independently over these exchanges projects Eq. (115) onto

$$C_{\text{Bose}}^{\text{Box}} = A_0(\tilde{\nu}, r) + \mathcal{O}((k_s/k_i)). \quad (116)$$

Hence the leading, fully Bose-symmetrized top-box clock signal is isotropic. The angular terms in Eq. (115) belong to an ordered box and must not be confused with the final identical-scalar angular dependence.

The scale dependence in the gauge-boson convention is

$$k_s^3 \left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2i\tilde{\nu}} + \text{c.c.} \sim k_s^3 \cos \left[2\tilde{\nu} \log \left(\frac{k_s^2}{4k_{12}k_{34}} \right) + \varphi \right]. \quad (117)$$

4 Independent checks

We close with checks that do not reuse the factorized derivation.

First, substituting Eq. (4) directly into the Dirac action reproduces both the mass term and the Yukawa vertex in Eq. (6). The transformation is unitary, so its Grassmann Jacobian is one. The two Majorana determinants reproduce one Dirac determinant, confirming the N_c , rather than $2N_c$, normalization.

Second, the lower-index type-2/type-3 helicity sum is the invariant tensor ϵ . Raising one index gives the identity, proving the rotation independence and the SK equality used in the cut. This also shows directly that no loop-momentum angular integral was silently dropped in Eq. (29).

Third, enumerating the two vertex choices as four-bit words gives exactly sixteen entries. The propagator dictionary in Eq. (17) gives four entries in each of the AA , AN , NA , and NN classes, so no chirality flow is duplicated or omitted.

Fourth, the conformal loop formula gives $k_s^{3+4if\tilde{\nu}}$. The two hard subgraphs give

$$\left(\frac{k_{12}}{2} \right)^{-2if\tilde{\nu}} \left(\frac{k_{34}}{2} \right)^{-2if\tilde{\nu}}, \quad (118)$$

so Eq. (111) has the stated nonanalytic ratio and overall momentum dimension k^{-9} .

Fifth, Eqs. (80) and (91) were checked distributionally: $d \operatorname{sgn}(t)/dt = 2\delta(t)$ and the principal-value transform gives the same derivative. This independently fixes the signs of all four SK kernels.

Sixth, multiplying the four pairs of 2×2 matrices in the $\mathbf{k} \parallel \hat{\mathbf{z}}$ frame and then applying the $SU(2)$ rotations reproduces all four traces in Eqs. (48), (53), (57), and (62). This independently leaves only 1 and c_{13} in the symmetric sector. For the antisymmetric sector, direct Pauli-matrix multiplication gives Eqs. (101), (102), and (103), including the opposite signs of the two triple-product terms. Pairing every word with its charge conjugate then eliminates that triple product. The tensor identity $2\mathcal{J}_T + \mathcal{J}_L = \mathcal{J}_0$ follows by tracing Eq. (88); it also follows independently from $\hat{\mathbf{q}} \cdot \hat{\mathbf{K}} = (q^2 + K^2 - k_s^2)/(2qK)$.

Seventh, multiplying the two late-time soft factors directly gives

$$(4q^2\tau_2\tau_3)^{if\tilde{\nu}}(4K^2\tau_4\tau_1)^{if\tilde{\nu}} = 2^{4if\tilde{\nu}}q^{2if\tilde{\nu}}K^{2if\tilde{\nu}}\prod_{j=1}^4(-\tau_j)^{if\tilde{\nu}}. \quad (119)$$

This verifies independently the separation into the bubble subgraph and the four extra time powers in Eqs. (27) and (89).

Finally,

$$|\kappa_+(\tilde{\nu})| = \left| \frac{\Gamma(-2i\tilde{\nu})}{\Gamma(-i\tilde{\nu})} \right|^2 = \frac{1}{4 \cosh(\pi\tilde{\nu})}, \quad (120)$$

so the two cut soft lines contain the expected large-mass suppression $|\kappa_+|^2 \sim e^{-2\pi\tilde{\nu}}/4$. Moreover $\kappa_- = \kappa_+^*$, $\mathcal{Q}_- = \mathcal{Q}_+^*$, and $f_0(i\tilde{\nu}, i\tilde{\nu}) = f_0(-i\tilde{\nu}, -i\tilde{\nu})^*$, which proves that the $f = -$ branch is the complex conjugate displayed in Eq. (111).

As a final seed check, setting $\tilde{\lambda} = 0$ and $\tilde{M}_R = \tilde{\nu}$ in Eqs. (2.52), (2.54), and (2.58) of Ref. [2] gives Eqs. (67)–(69) term by term. The reversal and complex-conjugation identities in Eq. (66) then reconstruct the $\bar{D}$ and $\check{D}$ blocks, so the four symmetric diagrams require no seed beyond the three functions $\mathfrak{J}$, $\mathfrak{J}$, and $\mathfrak{K}$. Expanding the energy deformation as in Eq. (94) proves the same statement term by term for all twelve nonzero antisymmetric diagrams.

A Closed form of the hard fermion seed

For completeness, this appendix collects the type-1, type-2, and type-3 hard seeds at zero chemical potential, using the integral formulae of Ref. [2]. They make the four surviving terms in Eq. (75) explicit in Gamma and generalized hypergeometric functions. The antisymmetric contribution remains in the exact principal-value form of Eq. (99).

For $\mathbf{k} \parallel \hat{\mathbf{z}}$, write

$$\mathcal{I}_{++}^{p_1 p_2} = \begin{pmatrix} A_1^{p_1 p_2} - B_2^{p_1 p_2} & 0 \\ 0 & A_2^{p_1 p_2} - B_1^{p_1 p_2} \end{pmatrix}, \quad (121)$$

$$\mathcal{I}_{--}^{p_1 p_2} = \left(\mathcal{I}_{++}^{p_2^* p_1^*} \right)^\dagger, \quad (122)$$

$$\mathcal{I}_{-+}^{p_1 p_2} = \begin{pmatrix} U_1^{p_1 p_2} & 0 \\ 0 & U_2^{p_1 p_2} \end{pmatrix}, \quad \mathcal{I}_{+-}^{p_1 p_2} = \begin{pmatrix} -V_2^{p_1 p_2} & 0 \\ 0 & -V_1^{p_1 p_2} \end{pmatrix}. \quad (123)$$

The entries are

$$A_1^{p_1 p_2} = \tilde{\nu}^2 \mathfrak{I}^{p_1 p_2}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, i\tilde{\nu}), \quad (124)$$

$$A_2^{p_1 p_2} = \mathfrak{I}^{p_1 p_2}(i\tilde{\nu}, -i\tilde{\nu}, 1 + i\tilde{\nu}), \quad (125)$$

$$B_1^{p_1 p_2} = \mathfrak{I}^{p_2 p_1}(i\tilde{\nu}, -i\tilde{\nu}, 1 + i\tilde{\nu}), \quad (126)$$

$$B_2^{p_1 p_2} = \tilde{\nu}^2 \mathfrak{I}^{p_2 p_1}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, i\tilde{\nu}), \quad (127)$$

$$U_1^{p_1 p_2} = \tilde{\nu}^2 \mathfrak{K}^{p_1 p_2}(2, 2), \quad U_2^{p_1 p_2} = \mathfrak{K}^{p_1 p_2}(1, 1), \quad (128)$$

$$V_1^{p_1 p_2} = e^{-i\pi(p_1 - p_2)} \mathfrak{K}^{p_1 p_2}(1, 1), \quad V_2^{p_1 p_2} = \tilde{\nu}^2 e^{-i\pi(p_1 - p_2)} \mathfrak{K}^{p_1 p_2}(2, 2). \quad (129)$$

The factorized seed is

$$\mathfrak{K}^{p_1 p_2}(x, y) = e^{\frac{i\pi}{2}(p_1 - p_2)} \frac{\Gamma(1 + p_1 - i\tilde{\nu})\Gamma(1 + p_1 + i\tilde{\nu})\Gamma(1 + p_2 - i\tilde{\nu})\Gamma(1 + p_2 + i\tilde{\nu})}{2^{2+p_1+p_2}\Gamma(p_1 + x)\Gamma(p_2 + y)}. \quad (130)$$

The nested seed is

$$\begin{aligned} \mathfrak{I}^{p_1 p_2}(x, y, z) &= i 2^{-2-p_1-p_2} e^{-\frac{i\pi}{2}(p_1+p_2)} \pi \operatorname{csch}(2\pi\tilde{\nu}) \Gamma(2 + p_1 + p_2) \\ &\times \left[\frac{e^{-\pi\tilde{\nu}}}{\Gamma(x)} \Gamma(1 + p_1 - i\tilde{\nu}) \Gamma(2 + p_1 + p_2 - 2i\tilde{\nu}) \right. \\ &\times \left. {}_4\tilde{F}_3 \left(\begin{matrix} 2 + p_1 + p_2, & 1 + p_1 - i\tilde{\nu}, & y, & 2 + p_1 + p_2 - 2i\tilde{\nu} \\ 2 + p_1 - i\tilde{\nu}, & 3 + p_1 + p_2 - z, & 1 - 2i\tilde{\nu} \end{matrix}; -1 \right) - (\tilde{\nu} \rightarrow -\tilde{\nu}) \right]. \quad (131) \end{aligned}$$

Here ${}_4\tilde{F}_3$ denotes the regularized generalized hypergeometric function. Equations (123)–(131), inserted into the $rs = N\bar{N}$ case of Eq. (37) with $p_+ = -1 + i\tilde{\nu}$, determine q_0 and q_1 as defined in Eq. (50), without any remaining time integral. The reversed $rs = \bar{N}N$ block follows from Eq. (18).

For the type-2 hard propagator, the corresponding lower-index matrices are

$$\hat{\mathcal{I}}_{++}^{p_1 p_2} = \begin{pmatrix} 0 & S_1^{p_1 p_2} + S_2^{p_2 p_1} \\ -(S_1^{p_2 p_1} + S_2^{p_1 p_2}) & 0 \end{pmatrix}, \quad (132)$$

$$\hat{\mathcal{I}}_{--}^{p_1 p_2} = \begin{pmatrix} 0 & S_3^{p_1 p_2} + S_4^{p_2 p_1} \\ -(S_3^{p_2 p_1} + S_4^{p_1 p_2}) & 0 \end{pmatrix}, \quad (133)$$

$$\hat{\mathcal{I}}_{+-}^{p_1 p_2} = \begin{pmatrix} 0 & W_1^{p_1 p_2} \\ -W_2^{p_1 p_2} & 0 \end{pmatrix}, \quad \hat{\mathcal{I}}_{-+}^{p_1 p_2} = \begin{pmatrix} 0 & W_2^{p_2 p_1} \\ -W_1^{p_2 p_1} & 0 \end{pmatrix}. \quad (134)$$

At zero chemical potential their entries reduce to

$$S_1^{p_1 p_2} = \tilde{\nu} \mathfrak{J}^{p_1 p_2}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, 1 + i\tilde{\nu}), \quad (135)$$

$$S_2^{p_1 p_2} = \tilde{\nu} \mathfrak{J}^{p_1 p_2}(i\tilde{\nu}, -i\tilde{\nu}, i\tilde{\nu}), \quad (136)$$

$$S_3^{p_1 p_2} = \tilde{\nu} \mathfrak{J}^{p_2 p_1}(i\tilde{\nu}, -i\tilde{\nu}, i\tilde{\nu}), \quad (137)$$

$$S_4^{p_1 p_2} = \tilde{\nu} \mathfrak{J}^{p_2 p_1}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, 1 + i\tilde{\nu}), \quad (138)$$

$$W_1^{p_1 p_2} = \tilde{\nu} \mathfrak{R}^{p_1 p_2}(2, 1), \quad W_2^{p_1 p_2} = \tilde{\nu} \mathfrak{R}^{p_1 p_2}(1, 2). \quad (139)$$

The second nested function is

$$\begin{aligned} \mathfrak{J}^{p_1 p_2}(x, y, z) &= i 2^{-2-p_1-p_2} e^{\frac{i\pi}{2}(p_1+p_2)} \pi \operatorname{csch}(2\pi\tilde{\nu}) \Gamma(2+p_1+p_2) \\ &\times \left[\frac{e^{\pi\tilde{\nu}}}{\Gamma(x)} \Gamma(1+p_1-i\tilde{\nu}) \Gamma(2+p_1+p_2-2i\tilde{\nu}) \right. \\ &\times {}_4\tilde{F}_3 \left(\begin{matrix} 2+p_1+p_2, 1+p_1-i\tilde{\nu}, y, 2+p_1+p_2-2i\tilde{\nu} \\ 2+p_1-i\tilde{\nu}, 3+p_1+p_2-z, 1-2i\tilde{\nu} \end{matrix}; -1 \right) - (\tilde{\nu} \rightarrow -\tilde{\nu}) \Big]. \end{aligned} \quad (140)$$

The type-3 seed follows without another integration,

$$\check{\mathcal{I}}_{ab}^{p_1 p_2} = - \left(\hat{\mathcal{I}}_{-a, -b}^{p_1^* p_2^*} \right)^*. \quad (141)$$

Converting the anomalous seeds to mixed indices as specified after Eq. (34) and using Eqs. (134)–(141) in the corresponding $rs = NN$ and $rs = \bar{N}\bar{N}$ cases of Eq. (37) supplies the type-2 and type-3 hard blocks in Eq. (75). Thus all four entries of $\mathcal{G}_{AA}$ are explicit.

References

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